An-Najah National University
Machine Learning Course

Project Assignment Due Date: 1/5/2026

Notes:

1. The assignment should be submitted by the due date time (Late Assignments will not be accepted for any reason).
2. The assignments are individual effort and copying the assignment will be treated as a cheating attempt.

Using the ***Bank Marketing Dataset associated in the zip file***, implement machine learning models to classify ***poutcome*** into ***4*** distinct classes.

What you need to do here is:

1. Submit the complete ***python code*** (Jupyter Notebook).

2. Perform Data Exploration and Preprocessing:
   a. Analyze dataset size, features, and class distribution
   b. Identify ***categorical*** features and ***numerical*** features
   c. Apply appropriate ***encoding*** for categorical variables
   c. Perform feature scaling (standardization or normalization)
   d. Split data into training (80%) and testing (20%) sets using ***stratified*** sampling

3. Implement Unsupervised Learning (K-Means Clustering):
   a. Apply ***K-Means*** clustering to the dataset (drop the ***poutcome and deposit*** targets)
   b. Determine the optimal number of clusters using ***elbow*** method.
   c. Compare clustering results with actual class labels.
   e. Discuss **limitations** of unsupervised learning for this classification task.

4. Implement Supervised Learning Models to classify the ***poutcome*** target:
   a. Neural Network: Build a fully connected network with ***at least*** one hidden layer
   b. Ensemble Method: Implement either ***Random Forest OR Gradient Boosting***
   c. Use appropriate activation functions and loss functions

d. Perform Hyperparameter Tuning and **kfold *stratified*** cross validation:
   i. Use grid search to optimize key parameters
   ii. For Neural Network: Tune number of hidden layers, neurons, learning rate, batch size.
   iii. For Ensemble: Tune number of estimators, max depth, learning rate.
   iv. Report your tuned hyperparameter values.

5. Evaluate Models and Report Results:
   a. Calculate and report Accuracy for all models
   b. Generate and display Confusion Matrix
   c. Report Precision, Recall, and F1-Score for each class
   d. report macro and micro metrics (Precision, Recall, and F1-Score)
   e. Compare training vs. test set performance

6. Write a report about your experiments:
   a. Which model performed best? Explain why with supporting evidence
   b. How did hyperparameter tuning affect model performance? Show your experiments
   c. Analyze the confusion matrix - which classes were most often confused?
   d. What challenges did you encounter during the project?
   e. Propose potential improvements or alternative approaches

7. Create visualizations:
   a. Plot class distribution in the dataset
   b. Visualize confusion matrices for all models
   c. Create comparison charts showing model performance metrics

Grading Criteria:

- Data Preprocessing: 20%
- Model Implementation: 25%
- Evaluation and Metrics: 25%
- Analysis and Discussion: 20%
- Report Clarity and Visualizations: 10%

Total: 100% *All the best* :)